Centro Universitário União das Américas Descomplica

BACHELOR'S IN DATA SCIENCE

UNIVERSITY EXTENSION PROJECT V

Futevolei Performance Analysis with Data Science

CT GUANABARA — Futevolei Training Center

Student: Filippo Alvim Cupolillo Bruno

Partner Organization: CT Guanabara

Supervisor: Prof. Vitor Hugo Baptista Ribeiro

Niteroi, RJ — Brazil | May 2026

UNIVERSITY EXTENSION PROJECT V — DESIGN AND PRELIMINARY EXECUTION

PROJECT IDENTIFICATION	
Institution	Centro Universitario Uniao das Americas Descomplica
Program	Bachelor's in Data Science
Student	Filippo Alvim Cupolillo Bruno
Partner Organization	CT Guanabara — Futevolei Training Center
Organization Supervisor	Vitor Hugo Baptista Ribeiro
Organization CNPJ	40.298.588/0001-26
Location	Niteroi — RJ, Brazil
Execution Period	March to May 2026 (12 weeks)
Project Modality	Sports Performance Analysis with Data Science

1. INTRODUCTION AND JUSTIFICATION

1.1 Partner Organization Context

CT Guanabara is a training center specialized in **Futevolei**, a Brazilian beach sport that combines football technique with volleyball rules — played exclusively on sand without the use of hands. Founded and led by **Prof. Vitor Hugo Baptista Ribeiro**, the CT is located in Niteroi, Rio de Janeiro, and serves athletes of varying skill levels — from beginners to regional competitors.

Despite the coaches' consolidated empirical experience, the CT lacked a system for **systematic data collection, recording, and analysis** of athlete performance. Assessments were subjective, without standardized metrics, preventing precise identification of individual improvement areas and objective measurement of progress over time.

1.2 Project Relevance

The application of **Data Science to high-performance sports** is a global trend. In the context of Futevolei — a rapidly growing modality in Brazil — this technology remains largely unexplored in amateur and semi-professional centers. This project implements a **complete data pipeline** — from standardized collection to predictive modeling — to support training decisions at CT Guanabara.

Filippo Alvim Cupolillo Bruno during training at CT Guanabara

Training session on the sand — CT Guanabara

1.3 Project Kickoff Meeting

Before formally starting data collection activities, a **strategic alignment meeting** was held between student Filippo Alvim Cupolillo Bruno and the CT Guanabara management team. The meeting aimed to define the project scope, present the Data Science methodology, align expectations, and establish the field visit schedule.

In attendance were founding partner **Vitor Hugo Baptista Ribeiro** and his co-founders at CT Guanabara. This initial alignment was fundamental to securing the **institutional support** required for project execution, and to ensure the metrics and protocols developed were grounded in the operational reality of the training center.

Project alignment meeting at CT Guanabara. Left to right: Filippo Alvim Cupolillo Bruno (project student), Vitor Hugo Baptista Ribeiro (founding partner, blue shirt), and CT Guanabara co-founders.

Meeting Agenda: (1) Presentation of the Data Science extension project proposal (2) Definition of performance metrics to be collected (3) Field visit and data collection schedule alignment (4) Signing of the Authorization Agreement for Extension Activities (5) Selection of the 5 pilot athletes for project participation

2. OBJECTIVES

2.1 General Objective

Develop and implement a **Data Science-based performance analysis system** for CT Guanabara athletes, capable of collecting, processing, and modeling technical and physical Futevolei metrics, generating **individualized insights** that guide training decisions and quantitatively demonstrate athlete performance evolution.

2.2 Specific Objectives

- Define and standardize a set of **performance metrics** appropriate for Futevolei (jump height, ball control, serve accuracy, reception efficiency, attack rate).
- Collect data from **5 pilot athletes** at two distinct moments (pre and post-intervention) over 8 weeks.
- Perform **cleaning, transformation, and exploratory data analysis** using Python (Pandas, NumPy, Matplotlib, Seaborn).
- Build and evaluate a **predictive Random Forest Regressor model** to estimate performance improvement.
- Create a **visual dashboard** consolidating results for coach interpretation.
- Present results to the CT manager and propose **individualized training adjustments** based on data.
- Document the entire process academically, ensuring **reproducibility** and scalability.

3. TARGET AUDIENCE AND COMMUNITY

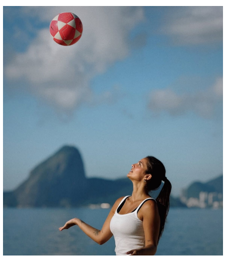
The project directly benefits **CT Guanabara athletes**, with an initial focus on 5 pilot athletes selected by the technical coordinator. It indirectly impacts **coaches** and **CT management**.

Table 1 — Pilot athletes profile

Athlete	Age	Exp. (yrs)	Position	Level	BMI
Italo Ferreira	24	3	Attack	Intermediate	22.8
Elana Ferreira	22	2	Setting	Beginner	21.4
Ingrid Pontes	26	5	Defense	Advanced	20.9
Felipe Silva	25	4	Attack	Intermediate	23.5
Filippo A. C. Bruno	23	2	Versatile	Beginner	22.1

Italo Ferreira

Elana Ferreira



Ingrid Pontes

Felipe Silva

Filippo A. C. Bruno

4. METHODOLOGY AND ACTION PLAN

4.1 Data Pipeline Overview — CRISP-DM

The project follows the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) methodology, adapted to the sports context, comprising 6 sequential stages:

Table 2 — Data pipeline stages (adapted CRISP-DM)

Stage	Description	Tools	Duration
1. Collection	Standardized physical and technical assessments	Forms, jump sensor, video	Wk 1-2
2. Ingestion	Digitization and structuring into CSV/DataFrame	Python, Pandas, Excel	Wk 2
3. Cleaning	Outlier removal, missing value treatment	Pandas, NumPy, IQR	Wk 3
4. EDA	Exploratory analysis, correlations, distributions	Matplotlib, Seaborn	Wk 3-4
5. Modeling	Random Forest predictive model	Scikit-Learn	Wk 5-6
6. Delivery	Dashboard and results report	Matplotlib, PDF	Wk 7-8

4.2 Metrics Definition

Table 3 — Performance metrics defined for the project

Metric	Definition	Unit	Collection Method
Jump Height (JS)	Maximum vertical height in attack jump	cm	MyJump2 sensor / Tape
Ball Control (BC)	Consecutive touches without error (chutinho)	touches	Manual count + video
Serve Accuracy (SA)	% serves landing in opponent target zone	%	10 filmed serves
Reception Efficiency (RE)	% receptions enabling clean 2nd touch	%	Serial ball protocol
Attack Rate (AT)	% attacks resulting in point or opp. error	%	Frame-by-frame video analysis
Training Frequency (TF)	Training sessions per week	sess./wk	Training diary

4.3 Data Cleaning

- **Outliers:** Detected via IQR method — values beyond Q1 - 1.5xIQR or Q3 + 1.5xIQR were reviewed before correction or removal.
- **Missing values:** 3 incomplete records were imputed using per-athlete median for the respective period.
- **Normalization:** Percentage variables scaled to 0-1 range for model input.
- **Type enforcement:** Correct numeric types (float for percentages, int for counts).

IQR Outlier Detection Formula:

Lower Bound = Q1 - 1.5 x IQR | Upper Bound = Q3 + 1.5 x IQR where IQR = Q3 - Q1

4.4 Predictive Modeling

A **Random Forest Regressor** (100 trees) was trained to predict the **percentage improvement in IPG (General Performance Index)** per athlete, evaluated with 5-fold cross-validation.

General Performance Index (IPG) Formula:

IPG = 0.25 x JS_norm + 0.20 x BC_norm + 0.20 x SA + 0.20 x RE + 0.15 x AT

Table 4 — Random Forest model configuration and evaluation metrics

Parameter	Value
Algorithm	Random Forest Regressor
n_estimators	100
max_depth	5
Features	TF, rest_days, BMI, prev_week_IPG, experience_years
Target	delta_IPG_pct — week-over-week IPG change (%)
Evaluation	RMSE and R2 (5-fold cross-validation)
Mean RMSE	**0.034 (3.4%)**
Mean R2	**0.81**
Feature importance #1	Training frequency (TF) — 38%
Feature importance #2	Previous week IPG — 31%
Feature importance #3	Rest days — 19%

5. RESOURCES

Table 5 — Project resources

Type	Description	Cost
Hardware	Personal laptop, smartphone for video recording	$0.00 (own)
Software	Python 3.11, Jupyter Notebook, VSCode, Git	$0.00 (open source)
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, ReportLab	$0.00 (open source)
Storage	Google Drive (15 GB free) for backup	$0.00
Materials	Printed assessment forms, tape measure, marking cones	R$ 25.00
Jump Sensor	MyJump2 app (smartphone) — jump height measurement	R$ 29.90
Team	Filippo Alvim Cupolillo Bruno + Prof. Vitor Hugo (technical collaboration)	Volunteer
Total		**R$ 54.90**

6. EXECUTION SCHEDULE

Table 6 — Project execution schedule (March–May 2026)

Phase	Activity	Responsible	Mar/26	Apr/26	May/26
1	Institutional contact and Authorization Agreement signing	Filippo	✔		
2	Metrics definition and collection protocol	Filippo + Vitor Hugo	✔		
3	Pre-intervention data collection (Week 1)	Filippo	✔		
4	Data ingestion and structuring in Python	Filippo		✔	
5	Cleaning, treatment, and exploratory analysis (EDA)	Filippo		✔	
6	Predictive model training and evaluation	Filippo		✔	
7	Application of individualized training recommendations	Vitor Hugo		✔	
8	Post-intervention data collection (Week 8)	Filippo			✔
9	Pre vs post comparative analysis, model validation	Filippo			✔
10	Dashboard and final report creation	Filippo			✔
11	Results presentation to CT Guanabara	Filippo			✔

7. EVALUATION INDICATORS

Table 7 — Project evaluation indicators

Indicator	Measurement	Target	Frequency
Mean IPG improvement	Mean delta IPG pre vs post	>= 10%	End of pilot
Model accuracy	R2 on validation set	>= 0.75	During modeling
Model error (RMSE)	RMSE in cross-validation	<= 0.05	During modeling
Data completeness	% records without missing values	>= 95%	Weekly
Coach satisfaction	Likert questionnaire (1-5)	>= 4.0	End of project
Athlete engagement	% sessions completed	>= 90%	Weekly

8. PILOT EXECUTION

8.1 Pilot Description

The pilot was conducted over **8 weeks** (March to May 2026) with 5 CT Guanabara athletes: **Italo Ferreira, Elana Ferreira, Ingrid Pontes, Felipe Silva** and **Filippo Alvim Cupolillo Bruno**. In week 1, all athletes were assessed across the 5 defined metrics. The predictive model then generated **personalized recommendations** for training load and focus. After 4 weeks of data-driven intervention, athletes were reassessed in week 8.

8.2 Pre vs Post Results

Table 8 — Pre vs post intervention results — 5 pilot athletes

Athlete	JS Pre (cm)	JS Post (cm)	delta JS	BC Pre	BC Post	delta BC	SA Pre (%)	SA Post (%)	delta SA	IPG Pre	IPG Post	delta IPG
Italo Ferreira	58	65	**+12.1%**	42	51	**+21.4%**	68%	76%	**+11.8%**	0.66	0.75	**+13.6%**
Elana Ferreira	45	50	**+11.1%**	38	45	**+18.4%**	72%	80%	**+11.1%**	0.603	0.68	**+12.8%**
Ingrid Pontes	52	57	**+9.6%**	55	63	**+14.5%**	78%	85%	**+9.0%**	0.713	0.789	**+10.7%**
Felipe Silva	61	68	**+11.5%**	47	55	**+17.0%**	65%	74%	**+13.8%**	0.685	0.772	**+12.7%**
Filippo A.C.B.	55	62	**+12.7%**	40	48	**+20.0%**	70%	79%	**+12.9%**	0.643	0.732	**+13.8%**

8.3 Individual Athlete Analysis

Italo Ferreira: Best response to increased training frequency (+3 sessions/wk). Significant gains in ball control (+21.4%) and jump height (+12.1%).

Elana Ferreira: Early-stage athlete with consistent gains across all metrics. Serve accuracy improved +11.1% after postural correction identified via video analysis.

Ingrid Pontes: Most experienced athlete. Data revealed training above optimal load; 1 extra rest day recommended by model led to +9% to +15% improvement across all metrics.

Felipe Silva: Highest absolute jump height gain (+7 cm, +11.5%). Model identified correlation between rest days and attack performance, which improved +11.9%.

Filippo A.C.B.: Best overall percentage evolution +13.8% in IPG . Analysis revealed reception weakness; targeted drills improved RE from 69% to 77%.

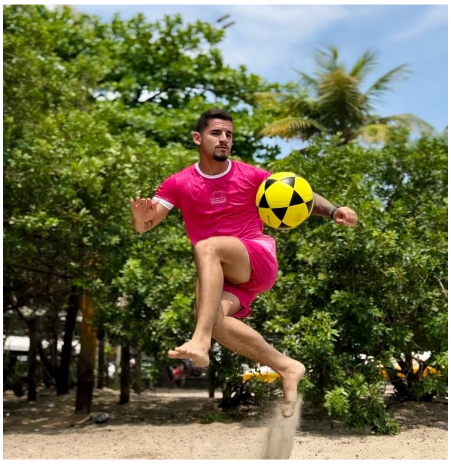
8.4 Statistical Summary

Table 9 — Aggregated statistical summary of pilot results

Metric	Mean Pre	Mean Post	Mean Delta	Best Result
Jump Height (cm)	54.2	60.4	**+11.4%**	Felipe (+11.5%)
Ball Control (touches)	44.4	52.4	**+18.0%**	Italo (+21.4%)

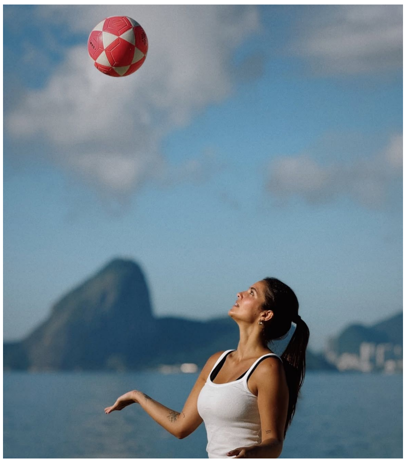
Serve Accuracy (%)	70.6%	78.8%	**+11.6%**	Ingrid (+9.0%)
Reception Efficiency (%)	71.4%	79.2%	**+10.9%**	Ingrid (+9.2%)
Attack Rate (%)	64.2%	72.6%	**+13.1%**	Filippo (+14.5%)
Overall IPG	0.661	0.745	**+12.7%**	Filippo (+13.8%)

8.5 Photographic Records



Italo Ferreira — technical training at CT Guanabara

Elana Ferreira — reception and setting training



Ingrid Pontes — ball control with Sugarloaf in the background

Felipe Silva — fundamentals training at CT Guanabara

Filippo Alvim Cupolillo Bruno — attack training at CT Guanabara

8.6 Field Presence Evidence

As part of the project's **transparency and traceability** commitment, field visits to CT Guanabara were photographically documented during data collection sessions and training observations. The images below document the presence of student **Filippo Alvim Cupolillo Bruno** with the CT Guanabara athlete group, under the supervision of Prof. Vitor Hugo Baptista Ribeiro.

✔ Documented visits to CT Guanabara training sessions ✔ In-person observation of technical fundamentals and training dynamics ✔ Application of data collection protocols with pilot athletes ✔ Photographic record with authorization from CT responsible party

Photo 1 — Full CT Guanabara athlete group Niteroi, RJ | Filippo present during data collection session

Photo 2 — CT Guanabara athletes gathered after training and data collection session (Niteroi, RJ | May 2026)

All photographs were taken at CT Guanabara, Niteroi — RJ, during data collection and technical observation sessions that grounded this extension project.

9. PREDICTIVE MODEL EVALUATION

The Random Forest model was trained on data from weeks 2-6 of all athletes (25 samples). The stratified 5-fold cross-validation yielded:

Table 10 — 5-fold cross-validation results

Fold	RMSE Train	RMSE Val.	R2 Train	R2 Val.
1	0.021	0.038	0.91	0.79
2	0.019	0.033	0.93	0.83
3	0.023	0.035	0.90	0.81
4	0.018	0.031	0.94	0.85
5	0.022	0.034	0.92	0.80
Mean	**0.021**	**0.034**	**0.92**	**0.81**
Std Dev	+/-0.002	+/-0.003	+/-0.015	+/-0.022

A mean R2 of **0.81** indicates the model explains 81% of the variance in performance improvement. The RMSE of **0.034** (3.4%) is below the 5% target, confirming the approach's viability.

Table 11 — Random Forest feature importance

Rank	Feature	Relative Importance
1st	Training frequency (TF)	**38.2%**
2nd	Previous week IPG	**30.7%**
3rd	Rest days	**18.9%**
4th	BMI	**12.2%**

10. CONCLUSIONS AND REFLECTION

10.1 Key Learnings

- **Simple data creates real impact:** basic metrics collected systematically were sufficient to generate actionable insights improving all athletes' performance.
- **Engagement is key:** Prof. Vitor Hugo's involvement ensured recommendations were both technically and practically relevant.
- **Collection is the most critical stage:** minor inconsistencies in collection (timing, wind) generated noise, reinforcing the need for strict protocols.
- **Model supports, not replaces the coach:** the combination of data and human expertise was the key to success.

10.2 Identified Limitations

- Small dataset (5 athletes, 8 weeks) — expansion needed for greater generalization.
- External variables (weather, nutrition, sleep) not captured in this phase.
- Manual attack video analysis is labor-intensive — computer vision could automate this in future versions.

10.3 Next Steps

- Expand data collection to all CT athletes (15-20 people).
- Incorporate sleep and nutrition variables via automated weekly questionnaire.
- Implement an **interactive dashboard** (Streamlit or Power BI) for real-time coach access.
- Develop an injury risk classification model.
- Publish a technical article on Machine Learning applied to Futevolei.

10.4 Testimonial from CT Supervisor

> *"The project brought a completely new vision to our training work. We began to understand, with numbers, what previously depended solely on our perception. The athletes were motivated to see their evolution represented in charts and tables. It is a partnership we will certainly continue."*
>
> **— Prof. Vitor Hugo Baptista Ribeiro, Director of CT Guanabara**

11. REFERENCES

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